

Environmental factors affecting horticultural crop production

(Effect of temperature, light, humidity, rainfall, wind and their stress on production and Measures to overcome environmental stress)

1. Temperature

- Effect:
 - High temperature can cause sunburn, poor fruit set, and increased transpiration.
 - Low temperature or frost can damage flowers, fruits, or even kill young plants.
- Stress Symptoms: Leaf scorching, fruit drop, delayed maturity.
- Mitigation:
 - Use of shade nets or greenhouses.
 - Mulching to regulate soil temperature.
 - Sprinkler irrigation during frost events.

2. Light (Sunlight)

- Effect:
 - Light affects photosynthesis, flowering, and fruiting.
 - Insufficient light leads to poor growth and lower yields.
- Stress Symptoms: Spindly growth, poor flower and fruit formation.
- Mitigation:
 - Proper plant spacing to ensure light penetration.
 - Supplemental lighting in controlled environments.

3. Humidity

- Effect:
 - High humidity favors fungal diseases and reduces transpiration.
 - Low humidity causes excessive water loss.
- Stress Symptoms: Leaf curling, disease outbreaks.
- Mitigation:
 - Ventilation in greenhouses.
 - Foggers or misting systems in dry conditions.

4. Rainfall

- Effect:
 - Excess rainfall leads to waterlogging and root rot.
 - Low rainfall causes drought stress.
- Stress Symptoms: Yellowing, wilting, nutrient leaching.

- Mitigation:
 - Drainage systems in heavy rainfall areas.
 - Drip irrigation and rainwater harvesting in dry areas.

5. Wind

- Effect:
 - Strong wind damages crops, increases transpiration, and spreads pests.
 - Stress Symptoms: Broken branches, lodging, desiccation.
 - Mitigation:
 - Windbreaks (e.g., tree rows or fences).
 - Staking or trellising of plants.
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General Measures to Overcome Environmental Stress

- Use of stress-tolerant varieties.
- Antitranspirants and growth regulators.
- Crop rotation and diversification.
- Proper timing of sowing and harvesting.
- Soil moisture conservation practices.

Question : Orchard establishment and management:

-Site selection and layout of orchard ; Planting, soil and water management practices, wind break and shelter belts ; Fertility and weed management

Orchard Establishment and Management

1. Site Selection and Layout of Orchard

- Site Selection:
 - Climate suitability (temperature, rainfall, frost risk)
 - Soil type: Deep, well-drained, fertile soils with pH 6–7.5
 - Topography: Prefer gentle slopes; avoid waterlogged or steep lands
 - Water availability: Year-round irrigation source is ideal
 - Accessibility to market and labor
- Orchard Layout:
 - Systems of planting:
 - *Square system*: Easiest for layout and interculture
 - *Rectangular system*: More efficient land use
 - *Hexagonal and Quincunx*: Accommodates more trees
 - *Contour system*: Used in hilly areas to reduce erosion
 - Layout includes roads, irrigation channels, windbreaks, and blocks of different crops

2. Planting, Soil and Water Management, Wind Breaks and Shelter Belts

- Planting Practices:
 - Best time: Dormant season (late winter to early spring)
 - Pits: Usually 1m x 1m x 1m filled with topsoil, compost, and neem cake
 - Spacing: Depends on species (e.g., Mango: 10m x 10m; Guava: 5m x 5m)
- Soil Management:
 - Regular tillage, cover cropping, and mulching
 - Use of organic matter and green manure to maintain fertility
- Water Management:
 - Use drip or basin irrigation for water efficiency
 - Ensure proper drainage to prevent waterlogging
 - Schedule irrigation based on soil moisture and crop needs
- Wind Breaks and Shelter Belts:
 - Rows of tall trees (e.g., Eucalyptus, Casuarina) around orchard
 - Protect from strong winds, reduce transpiration and physical damage

Wind Break: A row of trees or tall shrubs planted perpendicular to the prevailing wind direction to protect crops, soil, and infrastructure.

Purpose:

- Reduce wind speed
- Prevent soil erosion
- Minimize crop damage and moisture loss
- Improve microclimate for sensitive crops
- Example Trees: Casuarina, Eucalyptus, Poplar, Acacia

Shelterbelts: Multiple rows of trees and shrubs planted in a wider zone than windbreaks, usually in belt form.

Purpose:

- Broader protection from strong winds
- Control desertification and sand movement
- Create favorable conditions for crop production over a larger area
- Example: Used extensively in arid and semi-arid regions

3. Fertility and Weed Management

- Fertility Management:
 - Apply FYM/compdost before planting and annually
 - Balanced use of NPK fertilizers based on crop and soil testing
 - Use biofertilizers and micronutrients as needed
- Weed Management:
 - Manual weeding, mulching, and intercropping
 - Use of herbicides (with caution) like glyphosate around tree base
 - Maintain clean basins around trees

Question ? : Growth and development:

- Concept of growth and development, dormancy (seed and bud)
- Germination, juvenility and maturity
- Flowering, fruit set, fruit growth and development
- Fruit maturity and ripening, unfruitfulness, fruit drop - Tuber, rhizome and bulb development; Senescence

Growth and Development

1. Concept of Growth and Development

- Growth: Irreversible increase in size, volume, or weight due to cell division and enlargement.
- Development: Progression through various stages of life—germination to senescence, including physiological and biochemical changes.
- Growth and development are regulated by hormones and environmental conditions.

2. Dormancy (Seed and Bud)

- Seed Dormancy: Inability of viable seeds to germinate under favorable conditions.
 - Types: Physical, physiological, and morphological.
 - Breaking methods: Scarification, stratification, use of gibberellins.
- Bud Dormancy: Period when buds do not grow even under suitable conditions.
 - Common in temperate fruit crops (e.g., apple).
 - Broken by chilling (vernalization), pruning, or growth regulators.

3. Germination, Juvenility, and Maturity

- Germination: Activation of metabolic processes leading to seedling emergence.
 - Requires moisture, oxygen, temperature, and sometimes light.
- Juvenility: Phase when plant cannot flower or produce fruit; high vegetative growth.
- Maturity: Stage when plant acquires the ability to flower and bear fruit.

4. Flowering, Fruit Set, Fruit Growth and Development

- Flowering: Triggered by photoperiod, temperature, and hormonal balance.
- Fruit Set: Successful fertilization of ovule leading to fruit development.
 - Can be natural or induced via plant growth regulators (PGRs).
- Fruit Growth:
 - Occurs in three phases:
 1. Cell division
 2. Cell enlargement
 3. Maturation

5. Fruit Maturity and Ripening, Unfruitfulness, Fruit Drop

- Fruit Maturity: Stage when fruit is ready for harvest.
 - Physiological maturity: Maximum development.
 - Horticultural maturity: Based on consumer preference.
- Ripening: Softening, color change, sugar accumulation.
 - Regulated by ethylene in climacteric fruits (e.g., banana, mango).
- Unfruitfulness:
 - Causes: Lack of pollination, fertilization, or environmental stress.
 - Types: Structural, functional, or environmental.
 - Managed by cross-pollination, hormone sprays, or suitable cultivars.
- Fruit Drop:
 - May occur at early (natural thinning), middle (physiological), or late (pre-harvest) stages.
 - Controlled by NAA, 2,4-D, or improving pollination.

6. Tuber, Rhizome, and Bulb Development

- Tuber (e.g., Potato): Swollen underground stem storing starch.
 - Rhizome (e.g., Ginger): Horizontal underground stem for propagation.
 - Bulb (e.g., Onion): Compressed stem surrounded by fleshy leaf bases.
 - Their development is influenced by photoperiod, temperature, and nutrition.
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7. Senescence

- Natural aging process leading to cell and tissue death.
- Types: Whole plant, organ, or sequential senescence.
- Regulated by decreased cytokinin and increased ethylene/ABA.
- Can be delayed by using anti-senescent agents like cytokinins.

Question **?** : Basic principles of off season and protected horticulture and their prospects of Nepal

Basic Principles of Off-Season and Protected Horticulture

1. Off-Season Horticulture

- Definition: Cultivation of fruits, vegetables, and flowers during a period when they are not typically grown or available in the market.
- Principles:
 - Selection of early or late-maturing varieties
 - Use of climate-resilient practices
 - Adjusting planting and harvesting times
 - Use of microclimate modification (e.g., mulching, plastic tunnels)
- Examples:
 - Tomato and cucumber grown in winter using low tunnels
 - Off-season cauliflower, cabbage in higher altitudes

2. Protected Horticulture

- Definition: Growing crops under structures like greenhouses, polyhouses, net houses, or low tunnels to create a favorable microclimate.
- Principles:
 - Control of temperature, humidity, light, and airflow
 - Use of drip irrigation and fertigation
 - Integrated pest management inside structures
 - High-density planting for increased productivity

Prospects of Off-Season and Protected Horticulture in Nepal

Opportunities and Potential

- Agro-climatic diversity allows year-round production of different crops in different regions (Terai, mid-hills, and high hills).
- High market value for off-season vegetables and fruits, especially near urban centers (Kathmandu, Pokhara, Butwal).
- Government support through subsidies and training (e.g., PMAMP, ADS programs).
- Rising demand for pesticide-free, quality produce from protected environments.
- Can support youth employment, women farmers, and smallholder income.

Challenges

- High initial cost for protected structures
- Lack of technical knowledge and skilled manpower
- Limited access to quality planting materials and market linkages

Way Forward

- Promote low-cost structures like plastic tunnels and net houses.
 - Enhance extension services and farmer training.
 - Strengthen input supply chains and cold storage facilities.
 - Develop cooperatives for collective marketing of off-season produce.
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Concept of high density planting, multi-tier cropping, multiple cropping and agro forestry

1. High Density Planting (HDP)

- Definition: ***Growing fruit trees at closer spacing than traditional systems to maximize productivity per unit area.***
- Key Features:
 - Utilizes dwarf or semi-dwarf varieties
 - Requires pruning, training, and use of growth regulators
 - Common in crops like apple, mango, guava, banana, and citrus
- Advantages:
 - Higher yield per hectare
 - Early returns
 - Efficient resource use (light, space, nutrients)

2. Multi-Tier Cropping

- Definition: Cultivation of plants of different heights in the same area to maximize sunlight and space.
- Example:
 - Coconut (tall) + Banana (medium) + Pineapple (ground layer)
- Benefits:
 - Efficient use of vertical space
 - Increased overall productivity
 - Reduced weed growth and erosion

3. Multiple Cropping

- Definition: Growing two or more crops on the same land in one year.
- Types:
 - Intercropping: Growing two crops together (e.g., maize + beans)
 - Sequential cropping: Planting one crop after another (e.g., rice → mustard)
 - Mixed cropping: Growing different crops together without definite rows
- Advantages:
 - Better land utilization
 - Risk reduction due to crop diversity
 - Enhanced income

4. Agroforestry

- Definition: Integration of trees and shrubs with crops and/or livestock on the same land unit.
- Components:
 - Agricultural crops, forestry trees, fruit trees, and animals
- Examples:
 - Alley cropping (crops between tree rows)
 - Home gardens (diverse species in a household setting)
- Benefits:
 - Soil conservation
 - Biodiversity enhancement
 - Climate resilience and income diversification

Principles of urban and periurban horticulture; vermiculture, hydroponics and aeroponics and river bed farming

1. Principles of Urban and Peri-Urban Horticulture

Urban Horticulture

- Definition: Cultivation of vegetables, fruits, flowers, and herbs within urban areas (rooftops, balconies, community gardens).
- Principles:
 - Efficient space utilization (e.g., vertical gardening)
 - Use of container gardening and soil-less systems
 - Promotion of organic practices and recycling of kitchen waste
 - Ensuring local food security and aesthetic improvement

Peri-Urban Horticulture

- Definition: Horticultural production on the outskirts of cities, supplying fresh produce to urban markets.
- Principles:
 - Production of perishable crops (leafy vegetables, fruits)
 - Use of protected cultivation to increase yield
 - Efficient market linkage and transport system
 - Water-efficient technologies like drip irrigation

2. Vermiculture (Vermicomposting)

- Definition: Use of earthworms to convert organic waste into nutrient-rich compost.
- Principles:
 - Organic waste + moisture + shade = ideal worm activity
 - Improves soil structure, aeration, and nutrient content
 - Reduces urban waste and promotes circular farming

3. Hydroponics

- Definition: Growing plants without soil, using a nutrient-rich water solution.
- Principles:
 - Plant roots are suspended in or above the water/nutrient solution
 - Controlled environment (temperature, pH, nutrients)
 - Ideal for urban areas, rooftops, and space-limited settings

- Advantages: Faster growth, water-saving, no weeds or soil-borne diseases

4. Aeroponics

- Definition: Advanced form of hydroponics where plant roots are suspended in air and misted with nutrients.
- Principles:
 - Roots are exposed to high oxygen
 - Nutrient mist sprayed at intervals
 - Requires climate-controlled structures
- Benefits: Minimal water use, rapid growth, high space efficiency

5. Riverbed Farming

- Definition: Cultivation of short-duration crops on the sandy or fertile beds of rivers, especially during dry season.
- Principles:
 - Utilizes naturally moist, nutrient-rich riverbeds
 - Popular for cucumber, pumpkin, watermelon, bitter gourd, etc.
 - Allows off-season production and income generation
- Common in: Terai and riverbank areas of Nepal during the winter season

Fertilizer: Definition

Fertilizer is any natural or synthetic substance added to soil or plant tissues to supply essential nutrients required for plant growth.

Types of Fertilizers

1. Based on Nutrient Content

a) Macronutrient Fertilizers

- Supply major nutrients like Nitrogen (N), Phosphorus (P), Potassium (K)
- Examples:
 - Urea (N)
 - DAP – Diammonium Phosphate (N & P)
 - MOP – Muriate of Potash (K)

b) Micronutrient Fertilizers

- Supply trace elements: Zinc (Zn), Iron (Fe), Boron (B), Copper (Cu), Manganese (Mn)
 - Examples:
 - Zinc Sulphate, Borax, Ferrous Sulphate
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2. Based on Source

a) Organic Fertilizers

- Derived from natural sources like plants, animals, or microorganisms
- Examples:
 - Farmyard manure (FYM), compost, vermicompost, bone meal, oil cakes

b) Inorganic (Chemical) Fertilizers

- Synthetic or mined chemical compounds
 - Provide nutrients in readily available forms
 - Examples: Urea, DAP, SSP, MOP
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3. Based on Nutrient Composition

a) Straight Fertilizers

- Contain only one nutrient
- Example: Urea (46% N), SSP (16% P)

b) Complex Fertilizers

- Contain two or more nutrients, chemically combined
- Example: NPK (e.g., 10:26:26)

c) Mixed Fertilizers

- Physical mixture of two or more straight or complex fertilizers
- Customizable ratios

4. Specialty Fertilizers

- Slow/Controlled Release Fertilizers: Release nutrients gradually over time (e.g., sulfur-coated urea)
 - Water-Soluble Fertilizers: Fully soluble in water, used in fertigation
 - Liquid Fertilizers: Applied as foliar spray or via drip systems
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Classify horticultural plants based on growth habits and Photoperiodic response

1. Classification Based on Growth Habits

Growth Habit	Description	Examples
Herbs	Soft, non-woody plants, usually seasonal	Coriander, Lettuce, Basil
Shrubs	Woody plants, smaller than trees, with many branches	Rose, Hibiscus, Pomegranate
Trees	Large woody plants with a single main stem	Mango, Apple, Litchi
Climbers	Weak stems, climb using tendrils or support	Grapevine, Passionfruit, Cucumber
Creepers	Grow horizontally on ground	Watermelon, Pumpkin, Bottle gourd
Bulbous Plants	Store food in underground bulbs	Onion, Garlic, Tulip
Succulents	Store water in leaves/stems	Aloe vera, Cactus, Jade plant

2. Classification Based on Photoperiodic Response

Photoperiodism = Plant's response to length of day and night in relation to flowering.

Photoperiodic Type	Response to Light Duration	Examples
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Short Day Plants	Flower when day length is less than a critical period (8-10 hrs)	Chrysanthemum, Dahlia, Strawberry, Onion
Long Day Plants	Flower when day length is more than a critical period (More than 12-14) hrs	Lettuce, Spinach, Beetroot, Radish
Day Neutral Plants	Flower irrespective of day length (undefined 5 to 10 hrs)	Tomato, Cucumber, Rose, Brinjal

Apomixis

Definition:

Apomixis is a form of asexual reproduction in plants where seeds are produced without fertilization. It results in offspring that are genetically identical to the parent plant (clones).

Types of Apomixis

1. Adventive Embryony
 - Embryo arises directly from somatic cells (nucellus or integuments), not from egg cell.
 - Example: Mango, Citrus
 2. Apospory
 - Embryo sac forms from somatic cell of the ovule, not from meiosis.
 - Example: Some grasses, Hieracium
 3. Diplospory
 - Embryo sac develops from megaspore mother cell without meiosis.
 - Example: Dandelion
 4. Parthenogenesis
 - Embryo develops from an unfertilized egg cell.
 - Example: Some varieties of apple and pear
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Significance of Apomixis in Horticulture

- Maintains genetic purity of hybrids
- Helps in fixation of desirable traits
- Ensures uniformity in plant population
- Potential in seed production without pollination

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Micro propagation and mist propagation

1. Micropropagation

Definition:

Micropropagation is a method of rapid vegetative propagation using tissue culture techniques under sterile, controlled lab conditions.

Key Steps:

1. Explant selection (leaf, shoot tip, root, etc.)
2. Sterilization of explant
3. Culture initiation on nutrient medium (e.g., MS medium)
4. Shoot multiplication
5. Rooting
6. Hardening and transfer to soil

Advantages:

- Produces disease-free, true-to-type plants
- Rapid multiplication of elite or rare species
- Year-round production

Examples:

- Banana, Orchids, Potato, Gerbera, Strawberry
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2. Mist Propagation

Definition:

Mist propagation is a vegetative propagation technique where plant cuttings are rooted under intermittent mist (fine water spray) to maintain high humidity.

Principle:

Keeps humidity high and leaf temperature low to reduce transpiration and prevent wilting of cuttings.

Components:

- Propagation chamber or structure
- Mist nozzles and timer
- Rooting medium (e.g., sand, cocopeat)

Advantages:

- Promotes faster and successful rooting of softwood and leafy cuttings
- Saves water and labor
- Common in nurseries

Examples:

- Rose, Bougainvillea, Coleus, Tea, Rubber
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Write shortly on waves of fruit drop.**What are the measures to overcome this problem?****Waves of Fruit Drop**

Fruit drop in plants usually occurs in three main waves:

1. Early Drop (Post-Flowering Drop):
 - Happens soon after flowering due to unfertilized or weak flowers.
 - Common in apple, mango.
 2. June Drop (Physiological Drop):
 - Occurs due to nutrient or hormonal imbalance, usually when fruits are pea-sized.
 - Linked to competition for food among developing fruits.
 3. Pre-Harvest Drop:
 - Mature fruits fall just before harvest, often due to ethylene production or environmental stress.
 - Seen in citrus, apple.
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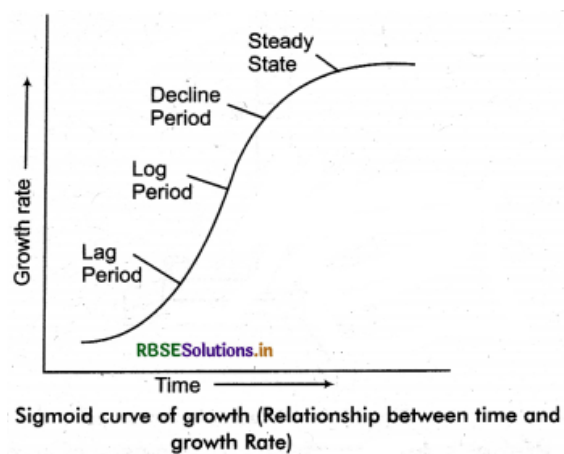
Measures to Overcome Fruit Drop

- Proper pollination and fertilization
- Balanced fertilization and irrigation
- Use of plant growth regulators (e.g., NAA – Naphthalene Acetic Acid)

- Thinning of excess fruits to reduce competition
- Pest and disease control
- Windbreaks to reduce mechanical drop

What is growth curve? How does growth differ from development?

Growth Curve: A growth curve is a graphical representation showing the increase in size or weight of a plant or plant part over time.



Typical Growth Curve

- S-shaped or sigmoid curve with four phases:
 1. Lag phase – Slow initial growth
 2. Log phase – Rapid growth
 3. Decelerating phase – Growth slows down
 4. Stationary phase – Growth stops or stabilizes

Difference Between Growth and Development

Aspect	Growth	Development
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Definition	Irreversible increase in size, weight, or volume	Progression of plant through different life stages
Nature	Quantitative (can be measured)	Qualitative (change in function/form)
Example	Increase in height or leaf area	Flowering, fruiting, senescence
Involves	Cell division, enlargement, elongation	Differentiation, maturation, organ formation

How do roots get initiated in air layering? Why is sphagnum moss most suited for air layering?

Root Initiation in Air Layering

In air layering, roots are initiated from a stem portion while it is still attached to the parent plant. Here's how it works:

- A ring of bark (girdle) is removed from the selected branch.
- This disrupts the downward flow of food (photosynthates), causing it to accumulate at the girdled site.
- The accumulated nutrients and plant hormones (especially auxins) stimulate root formation.
- The exposed area is then covered with a moist rooting medium like sphagnum moss and wrapped with plastic to retain moisture.

Why Sphagnum Moss is Best for Air Layering

- Excellent moisture retention: Keeps the rooting area consistently moist, essential for root initiation.
- Aeration: Provides enough oxygen around the developing roots.
- Sterility: Naturally resists decay and microbial growth, reducing the chance of infection.

- Lightweight and easy to apply: Easily molds around the stem and holds shape when wrapped.

Define training and pruning. Which pruning is best for vine plant ? Which kinds of training is ideal for grapes?

Training and Pruning: Definitions

- **Training:**
The process of giving shape or direction to a plant during its early growth to develop a strong framework and desired form.
 - **Pruning:**
The removal of unwanted, dead, diseased, or excess plant parts like shoots, branches, or roots to improve plant health, growth, and productivity.
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Best Pruning for Vine Plants

- For vine plants like grapes, spur pruning or cane pruning is best depending on the variety:
 - Spur pruning: Ideal for varieties that bear fruit on short shoots (spurs).
 - Cane pruning: Suitable for varieties that fruit on new long canes.
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Ideal Training System for Grapes

1. **Bower or Pergola System**
 - Most suitable for vigorous grape varieties.
 - Allows vines to spread over a horizontal wire frame.
 - Promotes good aeration, sunlight exposure, and higher yields.
2. **Kniffin System**
 - Two or more horizontal wires stretched between posts.
 - Shoots hang down freely. Suitable for moderately vigorous vines.
3. **Head System**
 - Plant is trained into a bush-like shape.
 - Used in dry regions or for low-vigor varieties.

Plumule and Radicle

These are the two main parts of a plant embryo in a seed:

1. Plumule

- Definition: The embryonic shoot of the plant.
 - Function: Develops into the stem, leaves, and upper parts of the plant.
 - Location: Arises from the tip of the embryonic axis, above the cotyledons.
 - Grows upward toward light (negative geotropism).
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2. Radicle

- Definition: The embryonic root of the plant.
 - Function: Develops into the primary root and root system.
 - Location: Found at the lower end of the embryo.
 - Grows downward into the soil (positive geotropism).
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Key Difference:

Feature	Plumule	Radicle
Part of Plant	Embryonic shoot	Embryonic root
Growth	Upward (toward light)	Downward (into soil)
Develops Into	Stem, leaves	Root system

Bulbs and Corms

Both bulbs and corms are underground storage organs that help plants survive unfavorable conditions and regrow each season, but they differ in structure.

1. Bulbs

- Structure: Consist of fleshy, layered scales (modified leaves) attached to a short stem (basal plate).
- Function: Store food in leaf bases.
- Examples: Onion, Garlic, Tulip, Lily

Features:

- Tunicated bulbs: Have papery outer covering (e.g., onion)
- Scaly bulbs: No papery covering (e.g., lily)

2. Corms

- Structure: A solid, swollen stem base with a tunic (dry covering) and nodes.
- Function: Store food in the stem tissue.
- Examples: Gladiolus, Crocus, Saffron, Taro (Colocasia)

Features:

- Solid inside (unlike layered bulbs)
- Have buds and nodes on the surface

Key Differences

Feature	Bulb	Corm
Type	Modified leaf	Modified stem
Structure	Layered scales	Solid, no layers
Storage organ	Leaf bases	Stem tissue
Examples	Onion, Garlic, Lily	Gladiolus, Crocus, Taro

Classification of Horticultural Plants Based on Non-Edible Parts

Horticultural plants can be classified based on which non-edible parts are used for ornamental, medicinal, aromatic, or other purposes, rather than for food.

1. Ornamental Plants

Grown for decorative purposes (flowers, foliage, form, or color)

- Examples:
 - Rose (flower)
 - Croton (colorful leaves)
 - Bougainvillea (bracts)
 - Ficus (foliage)

2. Medicinal Plants

Non-edible parts used in traditional and modern medicine

- Examples:
 - Neem – Leaves and bark
 - Aloe vera – Gel from leaves
 - Turmeric – Rhizome
 - Ashwagandha – Roots

3. Aromatic Plants

Non-edible parts used for essential oils, perfumes, cosmetics

- Examples:
 - Lavender – Flowers
 - Lemongrass – Leaves
 - Rosemary – Leaves and stems
 - Jasmine – Flowers

4. Timber and Utility Plants

Non-edible parts used for wood, fiber, or shelter

- Examples:
 - Bamboo – Stems
 - Teak – Wood
 - Mulberry – Leaves (for silkworms)

Sunken and Raised Bed Nursery

These are two types of nursery bed designs used in horticulture for propagating plants, each with its own advantages depending on soil type, water management, and environmental conditions.

1. Sunken Bed Nursery

- Definition: In a sunken bed, the nursery bed is prepared below ground level, often dug into the soil to a depth of 10-15 cm.
 - Features:
 - Water naturally drains downwards, preventing waterlogging.
 - The lower level helps in maintaining a stable root environment.
 - Often used for climate control, as it allows the bed to stay cooler in hot climates.
 - Advantages:
 - Improved water retention in dry climates.
 - Less risk of waterlogging in poorly drained soils.
 - Allows roots to grow deeper and more robust.
 - Disadvantages:
 - Soil erosion can be a problem in some areas.
 - Not suitable for areas with high water tables or flooding risk.
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2. Raised Bed Nursery

- Definition: In a raised bed, the nursery bed is elevated above ground level, typically built with soil, compost, or other growing media piled on top of the soil.
- Features:
 - Heightened bed improves drainage and provides easier access for planting and maintenance.
 - Often used in areas with poor or compacted soil or for better control of soil quality.
- Advantages:
 - Improved drainage, reducing risk of waterlogging.
 - Easier accessibility for workers, reducing strain.
 - Can be managed with controlled soil amendments and better aeration.
- Disadvantages:
 - More labor-intensive and requires more materials to construct.
 - Can dry out faster, requiring more frequent irrigation.

Comparison

Aspect	Sunken Bed Nursery	Raised Bed Nursery
Location	Below ground level	Above ground level
Water Management	Better water retention, less risk of waterlogging	Improved drainage, more frequent irrigation needed
Soil Type	Suitable for poorly drained soils	Suitable for compacted or poor soils
Labor Intensity	Lower cost, but can be difficult to access	Requires more materials, easier to work with
Climate Suitability	Better in hot, dry climates	Suitable in wet or cold climates

Orthodox Seeds and Recalcitrant Seeds

Seeds are classified based on their storage behavior, particularly their tolerance to drying and temperature.

1. Orthodox Seeds

- Definition: Seeds that can be dried to low moisture levels (5–7%) and stored for long periods under low temperatures without losing viability.
- Characteristics:
 - Tolerant to desiccation (drying)
 - Can be stored for many years
 - Ideal for seed banks and long-term conservation
- Examples:
 - Wheat, Rice, Maize, Tomato, Mustard

2. Recalcitrant Seeds

- Definition: Seeds that cannot tolerate drying and lose viability quickly when stored under low moisture and temperature.
- Characteristics:
 - Sensitive to desiccation and chilling
 - Short shelf-life
 - Require immediate sowing or special storage conditions (high humidity, moderate temperature)
- Examples:
 - Mango, Coconut, Jackfruit, Rubber, Avocado

Key Differences

Feature	Orthodox Seeds	Recalcitrant Seeds
Tolerance to Drying	High	Low
Storage Temperature	Low temperature (e.g., 5°C or below)	Cannot be stored at low temperatures
Lifespan	Long	Short
Storage Method	Seed banks	Must be kept moist or sown immediately
Examples	Wheat, Tomato, Rice	Mango, Jackfruit, Cocoa

Difference Between Top Working and Double Working

Feature	Top Working	Double Working
Definition	A technique where the canopy of a mature tree is cut back and new scions of desired variety are grafted.	A grafting technique where an intermediate variety is grafted between the rootstock and scion.
Purpose	To change the variety of an existing tree without replanting.	To overcome incompatibility between rootstock and scion or combine advantages of both.
Used On	Mature or old fruit trees (e.g., mango, citrus).	When scion and rootstock are not directly compatible.
Grafting Method	Generally involves cleft or bark grafting.	Involves two grafting operations: rootstock to interstock, then interstock to scion.
Example	Converting a local mango tree into a hybrid variety.	Using guava variety A as interstock between variety B and rootstock C.

Summary:

- Top Working = Changing variety on mature trees.
- Double Working = Adding an interstock to aid compatibility or modify traits.

How Roots Get Initiated in Layering

In layering, roots are formed on a stem while it is still attached to the parent plant. Here's how root initiation occurs:

1. Accumulation of Hormones and Nutrients

- When a portion of the stem is bent and covered with moist soil or rooting medium, the flow of food and hormones (especially auxins) accumulates at that point.
- This stimulates root formation from dormant cells in the stem.

2. Injury or Girdling (optional step)

- Sometimes, the bark is wounded or removed (called girdling) to:
 - Disrupt downward flow of nutrients
 - Encourage cell division and root initiation

3. Favorable Conditions

- The moisture, darkness, and aeration in the soil or sphagnum moss at the layering point create ideal conditions for root development.

4. Development of Adventitious Roots

- Roots that form from non-root tissue are called adventitious roots.
- These start emerging from the stem within a few weeks to months, depending on the plant species.

Source and Sink Relationship in Horticultural Crops

The source–sink relationship refers to the movement of photosynthates (mainly sugars) from the site of production (source) to the site of utilization or storage (sink) in a plant.

1. Source

- Definition: Plant organs that produce or export photosynthates (mainly leaves).
- Function: Carry out photosynthesis and supply energy-rich compounds (like glucose).
- Examples: Mature green leaves, sometimes storage organs (tubers during sprouting).

2. Sink

- Definition: Organs that consume or store the assimilates.
- Function: Support growth, development, or storage.
- Examples:
 - Growing fruits, flowers, roots, tubers, bulbs
 - Developing seeds
 - Meristems and expanding leaves

Importance in Horticulture

- Yield Determination: Balanced source–sink interaction ensures optimal fruit size, quality, and yield.
- Fruit Development: Fruits act as strong sinks; if source strength is weak (e.g., defoliation), fruits may remain underdeveloped or drop.
- Cultural Practices:
 - Thinning fruits reduces sink load and enhances size and quality.
 - Pruning influences the balance between vegetative (source) and reproductive (sink) growth.
 - Fertilization and irrigation enhance source activity and sink strength.

Summary:

- Source = producer of food (leaves)
- Sink = user or storer of food (fruits, flowers, roots)
- Their balance is critical for fruit set, growth, and overall productivity.

Difference Between Notching and Nicking

Both notching and nicking are horticultural techniques used to regulate plant growth and promote bud development, especially in fruit trees and ornamentals. Here's how they differ:

Feature	Notching	Nicking
Definition	Removal of a small piece of bark just above a bud.	Removal of a thin slice of bark just below a bud.
Purpose	To promote bud sprouting by preventing downward flow of auxins.	To suppress unwanted bud growth by disturbing upward flow of nutrients.
Effect on Hormones	Stops auxin flow downward, allowing bud to grow.	Disrupts cytokinin and nutrient flow upward, inhibiting growth.
Used For	Encouraging branching or flower bud initiation.	Controlling unwanted shoot development.
Location on Plant	Done just above a dormant bud.	Done just below a bud or shoot.

Summary:

- Notching = Stimulates bud growth
- Nicking = Inhibits or controls bud growth

Commercial Use of Plant Growth Regulators (PGRs) in Horticulture

Example:

Use of Gibberellins (GA₃) in Grape Production

- Purpose: To increase berry size and improve cluster looseness in seedless grape varieties like *Thompson Seedless*.
- Method: GA₃ is sprayed at flowering and fruit set stages.
- Effect: Enhances cell elongation, resulting in larger, better-shaped berries and improved market value.

PGR	Commercial Use
Auxins (IAA, IBA, NAA)	Rooting of cuttings in nursery propagation
Cytokinins	Promoting shoot multiplication in micropropagation
Ethylene (Ethephon)	Inducing uniform ripening in mango, tomato, pineapple
Paclobutrazol	Used to control plant height and promote flowering in mango

Plant Growth Promoters, Inhibitors, and Reagent Substances

1. Growth Promoters

These are plant hormones that stimulate growth and development.

Hormone	Function	Examples of Use
Auxins	Cell elongation, root initiation	Rooting in cuttings (e.g., IBA, NAA)
Gibberellins (GA ₃)	Stem elongation, fruit development	Berry enlargement in grapes
Cytokinins	Cell division, delay senescence	Micropropagation, improving shelf-life
Brassinosteroids	Promote cell expansion and stress resistance	Enhancing yield under stress

2. Growth Inhibitors

These are substances that suppress or delay plant growth.

Hormone	Function	Examples of Use
Absciscic Acid (ABA)	Induces dormancy, closes stomata	Helps in drought tolerance
Ethylene (Ethephon)	Promotes ripening, leaf/fruit drop	Uniform ripening in tomato, mango

Maleic hydrazide	Inhibits sprouting	Storage of onions and potatoes
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3. Reagent Substances (Synthetic chemicals used in regulation)

Substance	Role	Example
Ethephon	Releases ethylene	Used for ripening fruits
Paclobutrazol	Retards growth, induces flowering	Mango flowering control
Chlormequat (CCC)	Growth retardant, reduces internode length	In ornamental and cereal crops
2,4-D	Synthetic auxin, used as herbicide	Controls broadleaf weeds
